Natural latex

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History of natural latex

Natural rubber latex is derived from the rubber tree, Hevea brasiliensis. The rubber tree is a tall softwood tree, a native of the Amazon River basin. Industry botanists have concentrated their efforts mostly on this species for optimising the harvesting of rubber. Altough natural rubber can also be found in the Opium Poppy, Dandeloins,...

The rubber tree is now cultivated in virtually all tropical regions of Asia, Africa and South America. The ideal rubber growing regions should be 8 degree North of Equator, 10 degree South of Equator, high temperature, altitude not beyond 400m and high humidity.

The world rubber industry began to develop in the 1800s, with the invention of the masticator and the vulcanization process. Identifying the inventor of the vulcanization process is complex. Charles Goodyear (1800–1860) is generally credited as the first to come up with the basic concept.

Demand for rubber grew rapidly with the invention of the solid and later the pneumatic rubber tire and the demand for rubber insulation by the electrical industry. A notable fact in the early 1900s was also the emergence of production by smallholders, which produced rubber as one amongst several crops. Despite its original commercial development as a plantation crop, by the mid-1930s natural rubber production was evenly split between estates and smallholdings.

The Russian scientist Sergei Vasiljevich Lebedev created the first rubber polymer synthesized from butadiene in 1910. This form of synthetic rubber provided the basis for the first large-scale commercial production, which occurred during World War I as a result of shortages of natural rubber. This early form of synthetic rubber was again replaced with natural rubber after the war ended, but investigations of synthetic rubber continued. The big breakthrough of synthetic latex started in 1940 when the Goodrich Company developed a new and cheaper version of synthetic rubber known as Ameripol. Ameripol made synthetic rubber production much more cost effective, helping to meet the country's needs during World War II.

The production of synthetic rubber in the United States expanded greatly during World War II, since the Axis powers controlled nearly all the world's limited supplies of natural rubber by mid-1942. Military trucks needed rubber for tires, and rubber was used in almost every other war machine. By the early 1960s synthetic rubbers had overtaken natural rubber in volume.

The supply of natural rubber hinges upon the interplay among several factors, including production capacity, underlying technological change, input and processing costs as well as price differential with synthetic rubber. During the last 40 years natural rubber production grew on average 3.4% a year.

Extracting natural latex

Latex is often described as the sap of the Hevea tree. This is not an accurate description. The sap runs deeper inside the tree, beneath the cambium. Latex runs in the latex ducts which are located in a layer immediately outside the cambium.

This highlights the skill of the tapper. If the cambium is cut, then the tree is damaged, because the cambium is where all the growth takes place. Too much damage to the cambium, and the tree stops growing and stops making latex.

A tapper starts the trek around the plantation before dawn. At each tree a sharp knife is used to shave off the thinnest possible layer from the intact section of bark. A diagonal incision is made in the bark, allowing the latex to exude. The cut must be neither too deep, nor too thick. Either will reduce the productive life of the tree.

This starts the latex flowing, and the tapper leaves a little cup underneath the cut. The tapper returns a few hours later and collects the latex in the cup. In ordinary circumstances, this latex will normally coagulate into a lump in the bottom of the cup, called 'cup lump.' If the plantation manager wants to make latex, then the tapper must add a stabilising agent to the cup. Usually this is ammonia, which prevents the latex from coagulating.

The standard method in tapping is for one person to tap between 200 - 300 trees in 3-4 hours. After the incision the same person then collects the tapped latex. Tapping and latex extraction is normally carried out by paid labourers (estates) or household wolk (smallholdings).

If land values were omitted, labour would constitute the largest single cost per kg in the field production of natural rubber. Although this cost varies considerably among countries (owing to differentials in wages and yields), it tends to be prominent in all contexts. According to data drawn from private records of production and marketing enterprises, management and labour represented over half total direct expenditure on estates and smallholdings in Malaysia in the early 1990s.

Yield of natural latex

The plants generally have 32 years of economic life but they may live up to 100 years or even more than that. The plantation would start its yield from the 6th year onwards and then have a for tapping useful life of some 25 years. Afterwards the rubber wood is used for furniture and wooden parts of household items which are exported throughout the world.

Amount of Latex per tree per day - 35 grams (0.035 kg)
Number of trees per Acre - 200 Trees
Therefore day production from one acre - 7 Kg (200 x 0.035)
Number of days of tapping per year - 160 days (Considering monsoon rains)
Therefore annual production per year from one acre - 1120 Kg (7x160)
Production of Field Latex from one tree per year - 5.6 Kg (35% DRC)
One Hectare = 2.5 Acres
Production per Hectare per one year - 2800 Kg (1120x2.5)

Types of natural latex
A distinction is drawn between the following forms of natural latex:

Liquid

Ammonia is added to the latex to keep it liquid, and this is concentrated either by creaming or centrifuging. The resulting concentrate can be transported as a liquid. Depending on the quantity of ammonia added, a distinction is made between high-ammonia latex (HA) with 0.60 - 0.80% ammonia and low-ammonia latex (LA) with 0.20 - 0.29% ammonia.

The latex can be transported to the countries in which it is processed as latex concentrate in liquid form (which is the case for the moulded foam industry). This is done in barrels or in tank containers.



Solid

Coagulation of the natural latex can be achieved by letting the latex coagulate in the collecting cups or through adding sodium hydrogen sulfite ($NaHSO_3$) or using formic acid or acetic acid. After milling and washing the products are dried. The method of processing and drying determines the type of the end product.

For example:
Air Dried Sheets (ADS) are manufactured in smoke free drying rooms.

Ribbed smoked Sheets (RSS) are smoked in smokehouses.

Technically Specified natural Rubber (TSR) is processed according to a number of grades which are laid down in technical specifications drafted by ISO and are manufactured using new processing techniques.

Origin of natural latex

The Amazon basin (Brazil) enjoyed the monopoly of rubber production in the 19th century. It was collected by the Indians. Its monopoly was challenged in the early years of 20th century by south East Asian countries – Malaya (Malaysia), Sri Lanka and Indonesia.

Today the whole of south and Central American countries accounts for slightly over 2% of the world's output. While Asia produces about 92% of world's rubber.

Although, synthetic rubber production has increased many fold since World War II, yet natural rubber production, acreage and global distribution is increasing rapidly. In 1970, natural rubber production was slightly over 3 million metric tons. During 2011, production of natural rubber was 11 million metric tons and area under rubber plantation was over 9 million hectares.

Thailand

Thailand ranks first in natural rubber production in the world. This southeast Asian country has monsoon climate, which is most suitable for vegetative growth of rubber plant. During 2011, Thailand produced 3,394 thousands metric tons natural rubber, contributing around 31% of the world's total natural rubber production.

Indonesia

Indonesia is the 2nd largest natural rubber producer in the world. During 2011, its rubber production was 2,982 thousands metric tons. Dutch people introduce rubber plantation in Indonesia at the end of the 19th century. Sumatra and Borneo (Kalimantan) are two main rubber producing Islands. Tropical and monsoon climate suitability has encouraged the rubber plantation in Indonesia. Indonesia shared about 27% rubber output of the world.

Malaysia

Malaysia ranks third in rubber production after Thailand and Indonesia. During 2011 rubber output was 996 thousands metric tons. The west coastal and piedmont zone of the Malay Peninsula and western part of Kalimantan (Malaysian part) is important for rubber estates. Malaysia, still exports large quantities of natural rubber.

India

Natural rubber plantation was introduced in India in 1880, when rubber was planted in Travancore and Malabar regions, but commercial plantation started at relatively later date of 1902. India's natural rubber production has been increasing steadily over the past decade. During 2011 India produced 890 thousands metric tons natural rubber, contributing 7% to the world production.

Vietnam

Vietnam has emerged as fifth largest natural rubber producer in the world. Its production of natural rubber during 2011 was 812 thousands metric tons. The area around the Ho "Chi Minh" city is of fundamental importance for rubber cultivation. Vietnam exports the bulk of her natural rubber output to international market.

China

China is another outstanding country among rubber producing countries and has emerged as 6th largest rubber producer. During 2011, it produced 707 thousands metric tons rubber. China's rubber plantations are located at the hills of the south-east and in the Yangtze basin and coastal belt. China is an emerging industrial nation, its rubber demand is very large, therefore it is deficit in rubber production.

Other big exporting countries are the Philippines, Nigeria, Cote-d-Ivore and Sri Lanka.

The use of natural latex

The use of latex is widespread, ranging from household to industrial products, entering the production stream at the intermediate stage of as final products. Tires and tubes are the largest consumers of rubber. Also here over the years a lot of replacement of natural by synthetic latex has been seen but still many aircraft tires and inner tubes are purely made of natural latex due to the high cost of certification for aircraft use of synthetic replacements.

The remaining 44% are taken up by the general rubber goods (GRG) sector. This sector contains significant uses of latex for door and window profiles, hoses, belts, matting, flooring and dampeners (antivibration mounts) for the automotive industry. Gloves (medical, household and industrial) and toy balloons are also large consumers for latex. Significant tonnage of latex is used as adhesives in many manufacturing industries and products, although the two most noticeable are the paper and the carpet industries. Latex is also commonly used to make rubber bands and pencil erasers. The latex foam bedding industry is only using a very small percentage of the worldwide produced natural latex.

China being the fastest growing economy in the world is also the world's largest natural rubber consuming country. At 3.603 thousand tonnes it consumes 33% of the total rubber consumed over the world. USA has replaced India in 2011, being the second largest consumer of rubber at 1.029 thousand tonnes, followed by India at 958 thousand tonnes and Japan at 765 tonnes. Other major consuming countries include Malaysia, Indonesia and Thailand.

Conclusion

We believe that the consumer preference for Natural materials is growing. Consumers are becoming more conscious of the impact on environment of a product when making buying decisions.

Natural Latex occupies a special place as a product derived from plantations that help environment protection through:

• Absorb Carbon Dioxide from the atmosphere.
• Contribute to reducing global warming.
• Prevents soil erosion
• Rubber trees/rubber plantations are considered as a Forest Cover plantation that qualifies for FSC certification.

Beside this the plantations help to maintain the socio economic balance by providing employment to rural population particularly to illiterate older women. The flexible working hour's best suit the women who can make out a living without disturbing the family needs.

Matresses made from natural rubber are from a renewable source helping though the recoltation of the rubber sap maintaining the environmental and social balance in these countries where cultivated.

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